

HUGO Phase-III - Partner Placement Evaluator Design

1 Purpose

1.1 BACKGROUND

HUGO (Hub Delivery Global Optimization) is currently used by Acquisition Teams to visualize historical shipment demand and partner locations for recruitment decisions. Launched in 2024, the tool has successfully improved partner recruitment outcomes across multiple marketplaces by providing data-driven insights through heatmaps of historical demand and active partner locations. However, the current version lacks automated analysis capabilities for network capacity and partner placement optimization.

Phase III focuses on building the Partner Placement Evaluator, which will enhance HUGO by providing systematic recommendations for partner recruitment. This new component will analyze historical volume patterns, existing network capacity, and geographical coverage to guide placement decisions. By incorporating multiple data points including channel-wise volume breakdown, 52-week volume trends, and jurisdiction-level metrics, the system will help users make informed decisions about new partner recruitment based on network capacity analysis and historical performance data.

1.2 PROBLEM STATEMENT

The current partner recruitment process relies heavily on manual analysis through Excel formulas and individual judgment, leading to potential inconsistencies in decision-making across regions. Acquisition Teams lack systematic tools to evaluate network capacity saturation and determine optimal partner placement locations.

Additionally, the absence of historical tracking for recruitment decisions and their outcomes limits our ability to learn from past experiences and improve future recommendations.

These limitations impact the efficiency of our network expansion and potentially lead to suboptimal capacity utilization. HUGO will deliver a standardized, data-driven approach to ensure consistent recruitment criteria across regions and improve the success rate of new partner onboarding.

We plan to achieve 80%

1.3 GLOSSARY

- **MGRS:** Military Grid Reference System is the geocoordinate standard used by NATO militaries for locating points on Earth. Locations on earth are represented using an alphanumeric string.
- **ZAP5:** ZAP is tailor made Amazon Postcodes that have improved awareness of demand/population distributions and adhere to barriers such as natural boundaries, road and rail network in order to provide a homogeneous delivery experience. These postcodes are designed as building blocks for efficient jurisdiction planning, both for regular station jurisdictions, as well as sub-jurisdictions for alternative delivery methods.
- **SPOZH:** (Shipments per On Zone Hours) is the average number of packages on a route divided by the on-zone hours per route, where on-zone hours is the sum of transit and service times. SPOZH is one of the ways to measure the last-mile productivity, hence, a proxy for the last-mile cost per package (LMCPP). This is used for most of the AMZL last mile expansion planning decisions.
- **Polygons:** Polygon is geo-shape that encompasses a geographical area.
- **Partitions:** Partitions denote the types of divisions that are made in the map for analysis. The partition types supported by HuGO are MGRS 100m, MGRS 1km, MGRS 10km, ZIP, ZAP5 and Jurisdiction.
- **Bounding Box:** Geographical rectangle defining map view boundaries
- **Partner Placement Evaluator:** Tool to assess optimal partner placement

- **ADV:** Average Daily Volume
- **BAU:** Business As Usual
- **Attainable Volume:** Expected package volume in a given recruitable area or for a partner
- **Go/NoGo Decision:** Recommendation for partner recruitment. Go = Partner can be recruited. NoGo = Partner should not be recruited.
- **Channel Type:** Delivery service category (Hub, AMZL, 3P)
- **Delivery Station Jurisdiction:** The transportation network area that is covered by the delivery station.
- **HSS:** Hub Search Service which provides comprehensive query parameters to search Hub Delivery partner data.

2 Requirements

2.1 FUNCTIONAL REQUIREMENTS

The functional requirements have been divided into 3 milestones:

1. **M0:** Data Readiness for Partner Placement Evaluator - Ensure all backend ingestion, APIs, and essential data are available for the Partner Placement Evaluator to function
2. **M1:** Partner Placement Evaluator - Deliver the UI for Partner Placement Evaluator integrated with data from M0.
3. **M2:** Insights and Visualization Enhancements - Provide enhanced visualization to support analysis, planning, and user experience beyond core recruitment workflow.

Detailed requirements can be found in the BRD: Hub Tech BRD 2025 | HUGO Phase III / IV

M0: Data Readiness for Partner Placement Evaluator

1. [Volume Breakdown] Enhance storage to have a breakdown of Volume channels - Enhance storage to have a breakdown of 3P, AMZL, and Hub Volume
2. [Volume Stats] Store 52-week volume stats - Calculate and store actual shipment volume 52-week high, low, and average for Total Package Volume (Total, AMZL channels and 3P carrier), Hub Delivery Attained Volume, and Hub Delivery Attainment (Total and AMZL channels) can be viewed in the information panels at the bottom of the screen.
3. [Partition-level Attainable Volume] Calculate and return attainable package volume and number of recommended partners for BAU and peak for jurisdiction
4. [Lead-level Attainable Volume] Estimate attainable package volume for a prospective partner
5. [Recommendation Audit] Record recommended go/no-go decision along with selected filters - Store the entered target ADVs and the partners that were recommended for Go/NoGo along with the selected filters and the forecasted ADV.

M1: Partner Placement Evaluator

1. [UI] Tab for Partner Placement Evaluator
 - a. Select and View a Recruitment Area
 - i. User shall be able to search by typing the zipcode, zapcode, or exclusive jurisdiction find their recruitment area, and center the map on the selected recruitment area with a view of its polygon overlaid on the map (default to be determined by country).
 - ii. User shall be prompted to select Target Hub Delivery Attainment percentage. Must be a whole integer (default to be determined by country)
 - iii. User shall be prompted to select the Partner Delivery Radius. Must be a whole integer in increments of 50m (default to be determined by country).

- iv. User shall be prompted to select Recruitment Target: 1) BAU Target ADV (52-week Average) or 2) Peak Target ADV (52-week Peak) (default to be determined by country)
 - v. User shall be prompted to select the cycle from a drop down menu (default to be determined by country).
 - vi. User shall be prompted to select the specific work days of the week (default to be determined by country).
 - vii. Once all of the recruitment parameters (above) are defined, the user can view a map of the Hub Delivery Recruitable Attainable Package Volume - Estimated areas within the selected recruitment area. The algorithm will consider all active and onboarding Hub Delivery partner as attained volume, and all Prospective Partner (New, Working, Prospect) and Benched Partner will be excluded until selected for a recruitment evaluation (Go / No Go).
 - viii. User shall be able to pan and zoom the map to view of the entire recruitment area, but not beyond a partner radius (selected by user) beyond the parameter of the recruitment area polygon.
 - ix. User can select each recruitable area partition.
 - x. User can identify the recruitable areas clearly and use a contrasting color from non-recruitable areas.
- b. View Number of Recommended Partners
- i. User will be able to select a recruitable area partition on the map and see in a side panel the following:
 - 1. Recommended number of partners to recruit based on the Recruitable Number of Recommended BAU Partners - Estimated
 - 2. Recommended number of BAU partners to recruit based on the Recruitable Number of Recommended **BAU** Partners - Estimated
 - 3. Recommended number of PEAK partners to recruit based on the Recruitable Number of Recommended **PEAK** Partners - Estimated
 - 4. Active Partners
 - 5. Onboarding Partners
 - 6. Prospective Partners by lead status (New, Working, Prospect)
 - ii. User shall be able to export a list of all filtered Prospective Partners within the highlighted recruitable area to a .CSV file, including Store Name, Address, Latitude, Longitude, Lead Status, and a clickable link to the store's Salesforce page.
- c. Select / Deselect Active and Prospective Partners
- i. User shall be able to click on a Hub Delivery Active and Onboarding Partner map pin and select an option to include or exclude them from the Recruitable Attainable area calculation.
 - ii. Map pin will change from default color to gray.
- d. Go / No Go Evaluation
- i. User shall be able to click on a prospective partner map pin and select an option to perform a go / no go evaluation.
 - ii. User shall have the option to edit the delivery radius and center point (Lat, Long).
 - iii. User accepts or rejects the go / no go decision.
 - iv. System keeps a record of the go / no go decision, estimated ADV, timestamp, who made the decision, and the parameters selected, including target attainment percentage,

M2: Insights and Visualization Enhancements

- 1. [Adhoc Lead] View an adhoc lead on the map - View an Ad Hoc Business Location(s) on the Heatmap
- 2. [Volume Breakdown] Update UI to have a breakdown of volume by channel type
- 3. [Volume Stats] Display 52-week stats on UI in the package volume tiles

4. [Partner Delivery Volume] Ingest partner delivery data to display partner performance
5. [Recommendation Audit] Update the recommendation periodically with the onboarding decision and achieved ADV
6. [Partner Radius] Ingest Partner Radius
7. [Partner Radius] UI Partition Overlay - Add Partner Delivery Radius to Partition Overlay Feature for the Heatmap
8. [Partner Radius] UI User-entered Partition Overlay
9. [Jurisdiction-level Data] Store Package Volume data for Jurisdictions
10. [Jurisdiction-level Data] Support Hub Delivery Jurisdiction in the heatmap

2.2 NON-FUNCTIONAL REQUIREMENTS

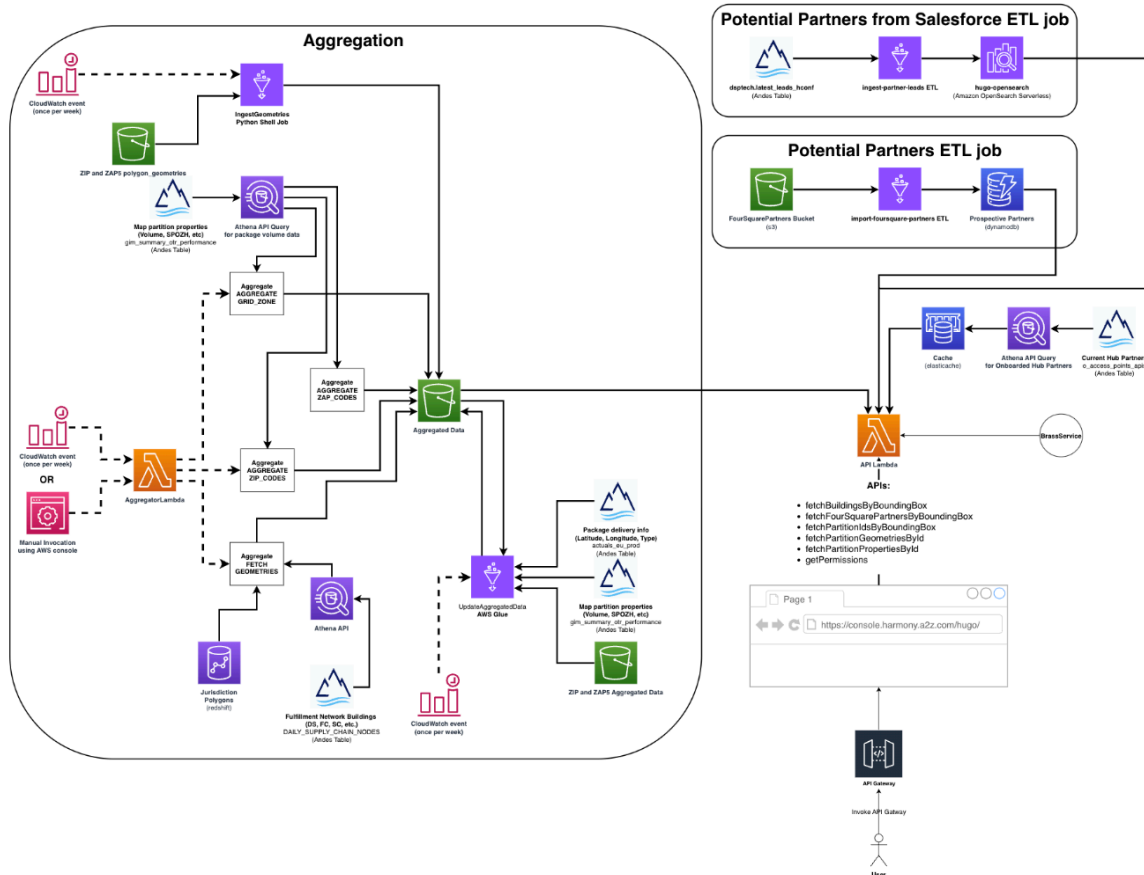
- System should maintain historical volume data with weekly granularity for 52-week calculations for quick retrieval
- Storage should support retrieval of aggregated metrics at both partition and partner level
- Design should allow for easy addition of new filtering criteria and metrics
- APIs should return filtered responses within acceptable latency for map rendering, limiting the need to perform client-level filtering
- System should validate incoming data completeness and accuracy
- Data anomalies should be flagged and logged for investigation
- APIs should be designed to accommodate additional filtering parameters from the client
- UI components should be modular for easy enhancement

2.3 OUT OF SCOPE

This design focuses specifically on the P0 requirements defined in Phase III BRD and Partner Placement Evaluator. It does not

- include the rendering and reporting of the recommendations provided by Hugo.
- take into account MCO.
- cover migration from Actuals to GDD

3 Current Architecture Overview



[Link](#)

HUGO is primarily divided into 2 major components:

1. Aggregation: Scheduled ETLs and aggregation jobs to pre-load the data from various sources
2. API: Lambda for fetching the required data from the UI, exposed via API Gateway.

More details here: [HUGO Architecture and Data Flow](#)

We already have jobs aggregating the following data:

1. Geographical Polygons - Geo-shape encompassing zones of different partition types, like ZIP, MGRS, Jurisdiction.
2. Package Volume Data - Data for total volume (AMZL + Hub + 3P), Hub Attained Volume, Hub Attained %, and SPOZH.
3. Hub Delivery Partners - Relevant partner fields like name, address, geocodes, and status.
4. Salesforce Leads - Relevant leads details like name, address, geocodes, and status.

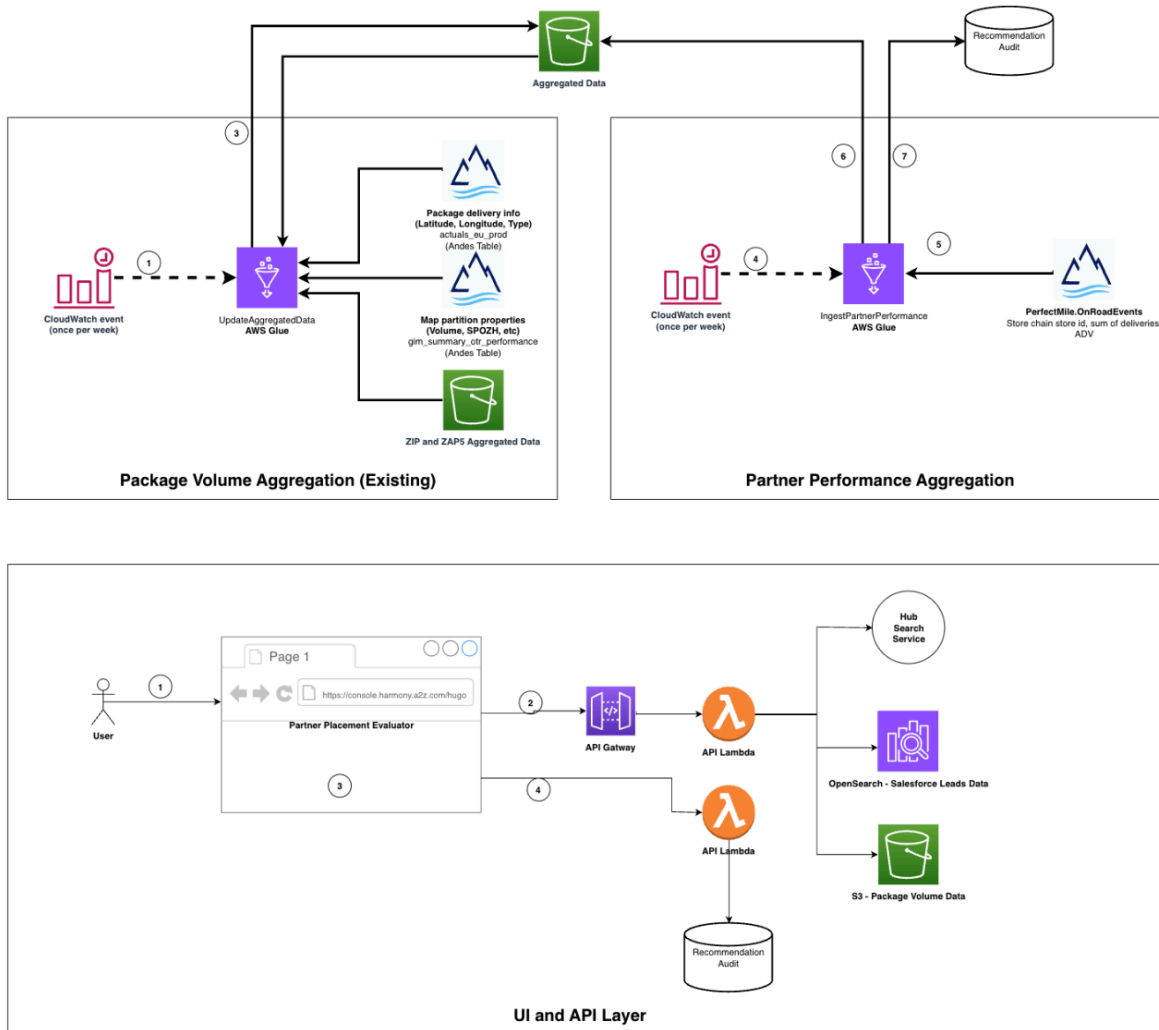
We also have APIs exposing this data to the frontend:

1. `GetPropertyIdsByBoundingBox`: Returns all partition ids for a given partition type
2. `GetGeometriesById`: Returns bounding boxes for all provided partition ids
3. `GetPropertiesByPartition`: Returns package volume data for provided partition ids
4. `GetProspectivePartners`: Returns list of leads/prospective partners for the input filter (status, bounding box, creation

date)

5. GetOnboardedPartners: Returns list of onboarded Hub Delivery partners for the input filter (status, bounding box, creation date)

4 High Level Design



Package Volume Aggregation

1. EventBridge schedule that runs the package data aggregation job weekly
2. Data read from Actuals, and GIM.summary_otr_performance tables and Hugo's ZIP and ZAP geometries bucket
 - a. **Update** - Read Max, Min, and Average using SQL queries along with cycle information
3. Update HUGO's aggregated data bucket with package volume data for the running week

Partner Performance Aggregation

4. EventBridge schedule that runs the partner data aggregation job weekly

5. ADV and delivered package volume for Hub deliveries at partner-level read from PerfectMile.OnRoadEvents table
6. Partner-level ADV stored in existing S3 aggregated data bucket
7. Partner-level ADV for HUGO-recommended partners stored in Recommendation audit table

UI and API Layer

1. User opens Partner Placement Evaluator on Hugo UI
2. Harmony app invokes API Lambda through API Gateway - multiple APIs to fetch polygon geometries, package volume heatmap, leads data, and partner data
3. Calculations made based on the user-provided input to get attainable volume
4. Calculations and recommendation provided by Hugo stored in RecommendationAudit storage

4.1 DATA INGESTION

4.1.1 Data Fields & Sources

	Existing Integration
	New Integration

	A	B	C	D	E	F
1	Category	Field	Is Already Ingested?	Source	Target	Is Personal Data?
2	Volume Data Fields	total_volume	Yes ▾	PerfectMile OnRoadEvents / Geocoding Data Daily	S3	No
3		hub_volume	Yes ▾		S3	No
4		amzl_volume	No ▾		S3	No
5		3p_volume	No ▾		S3	No
6		cycle	No ▾		S3	No
7		mgrs_id / zip / zap	Yes ▾		S3	No
8		jurisdiction id	No ▾		S3	No
9		week_number	Yes ▾	To be derived		No
10	52-Week Statistics	weekly_volume	Yes ▾			No
11		52_week_high	No ▾			No
12		52_week_low	No ▾			No
13		52_week_average	No ▾			No
14		calculation_date	No ▾			No
15	Partner Performance	partner_delivery_volume	No ▾	PerfectMile OnRoadEvents	S3	No
16		actual_adv	No ▾	PerfectMile OnRoadEvents	S3	No
17		target_adv	No ▾	HSS/APIS	ElastiCache	No
18		partner_capacity	No ▾	HSS/APIS	ElastiCache	No
19		days_worked	No ▾	HSS/APIS	ElastiCache	No
20		onboarding_date	Yes ▾	HSS/APIS	ElastiCache	No
21		id	Yes ▾	HSS/APIS	ElastiCache	No
22		delivery_radius	No ▾	HSS/APIS	ElastiCache	No
23		radius_unit	No ▾	HSS/APIS	ElastiCache	No
24		latitude	Yes ▾	HSS/APIS	ElastiCache	Yes
25		longitude	Yes ▾	HSS/APIS	ElastiCache	Yes
26	Jurisdiction Data	jurisdiction_id	No ▾	To be derived		No
27		is_hub_delivery_exclusive	No ▾			No
28		jurisdiction_volume	No ▾			No
29		jurisdiction_boundaries	No ▾			No
30	Recommendation Audit	entered_target_adv	No ▾	UI Input / To be derived		No
31		lead_record_id	No ▾			No
32		selected_filters	No ▾			No
33		forecasted_adv	No ▾			No
34		recommendation_timestamp	No ▾			No
35		recommendation_status (go/nogo)	No ▾			No
36		attainable_volume_bau	No ▾			No
37		attainable_volume_peak	No ▾			No
38		recommended_partner_count_bau	No ▾			No
39		recommended_partner_count_peak	No ▾			No
40		partner_estimated_volume	No ▾			No
41		delivery_radius	No ▾			No

4.1.2 Volume Data Fields & 52-Week Stats

The volume data is currently being ingested in the following format from GIM and Actuals tables (migration planned for GDD and PerfectMile.OnRoadEvents) for each partition type except Jurisdiction in S3:

Partition Id	Total Volume	Hub Attainment %	Hub Attained Volume	SPOZH
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Approach 1: Extend existing storage (S3) for additional fields (Recommended)

Here, we will extend the existing storage with volume breakdown - amzl volume, and 3p volume, while adding additional files for further categorization like cycle-level data and 52-week stats.

The data is queryable using SQL-like queries through S3Select.

Current Directory Structure per partition type (Directory - hugo-aggregated-data-region-stage/countryCode/aggregated-data-db/YYYY-W#):

- MGRS100, MGRS1K, MGRS10K, MGRS100K - *.csv (* = 100K MGRS grid ids)
- ZAP5 - ZAP5.csv
- ZIP - ZIP.csv

Proposed Directory Structure per partition type:

- MGRS100, MGRS1K, MGRS10K, MGRS100K (Sample URI - s3://hugo-aggregated-data-na-beta/US/aggregated-data-db/2025-W28/MGRS10K/02R.csv) -
 - *.csv (* = 100K MGRS grid ids)
 - stats.csv
- ZAP5 (Sample URI - s3://hugo-aggregated-data-na-beta/US/aggregated-data-db/2025-W28/ZAP5/ZAP5.csv) -
 - ZAP5.csv
 - stats.csv
- ZIP (Sample URI - s3://hugo-aggregated-data-na-beta/US/aggregated-data-db/2025-W28/ZIP/ZIP.csv) -
 - ZIP.csv
 - stats.csv

Proposed File Structure

Volume Breakdown -

Partition Id	Cycle	Total Volume	Hub Attainment %	Hub Attained Volume	SPOZH	Total AMZL Volume	3P Volume
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Stats -

Partition Id	Cycle	Mon-Avg	Mon-High	Mon-Low	Tues-Avg	Tues-High	Tues-Low
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The above file will be stored for each package volume breakdown - Total Volume, Hub Volume etc.

Pros:

1. Minimal changes to existing data pipeline
2. Backwards compatibility with current systems
3. Reduced development time for implementation
4. Easier to maintain data consistency with existing reports

Cons:

1. May not be optimized for new query patterns and might require complex queries to extract new data points
2. Could lead to performance issues as data scales and might require frequent management
3. Less flexibility for future expansions as we extend support for daily package volume, and breakdown by package type

Approach 2: Create new storage as per the latest requirements

We will design a dedicated storage layer (for example, new S3 bucket structure, or a purpose-built table store like Glue catalog table) for the enhanced schema.

Instead of extending the existing files, we create a clean structure optimized for the new breakdown fields (AMZL/3P split, cycle-level, 52-week stats, daily package breakdown).

Pros:

- Clean schema optimized for new requirements
- Flexible for future expansions and additional data breakdowns

Cons:

- Requires new data pipeline and migration effort for existing use cases
- No backward compatibility (need dual maintenance until migration complete)

4.1.3 Partner Details

Since multiple partner fields are already or will be available in HSS response, we will fetch the following fields ~~in real-time~~ from HSS (expected average latency < 1000 ms):

- store id
- target_adv
- operational days
- onboarding_date
- delivery_radius
- radius_unit
- latitude
- longitude

Why not extend the existing cache that reads from APIS data warehouse table?

We will migrate from APIS data warehouse to HSS, supporting real-time queries. This migration addresses two critical requirements:

1. Reduce UI latency from current daily country-level caching (>1 minute)
2. Access additional partner fields not available in APIS, including capacity and target ADV

Additional Requirements from HSS:

HSS currently does not support the below 2 queries/response fields which we will add -

- Return partner creation time in SearchHub response
- Support GeoSpatial query

4.1.4 Partner Performance

We will store the partner performance data, i.e., delivered volume in the existing S3 Volume Data storage, queryable by partner id.

This data will be rendered with partner details, including target ADV and delivery radius to render whether the partner performance is green (actual ADV < target ADV) or red (actual ADV >= target ADV).

PartnerId	Cycle	ADV	Packages Delivered
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4.1.5 Jurisdiction Data (TBD)

M2 requirement - requires calculation based on MGRS grid data.

4.2 API DESIGN

We will leverage existing APIs and add new fields for additional data. The frontend will make multiple API calls to gather required data in parallel, same as it does now for the heatmap view.

Existing APIs:

- GetPropertyIdsByBoundingBox: For jurisdiction/MGRS mapping
- GetGeometriesById: For polygon visualization
- GetPropertiesByPartition: For volume data
 - extended for returning volume breakdown by channel along with 52-week stats
 - extended to support jurisdiction-level volume data
- GetProspectivePartners: For lead information
 - extended for returning supported work days
- GetOnboardedPartners: For active partner data
 - extended for returning supported work days, radius, and delivery volume data

Approaches for a new API fulfilling remaining requirements documented below:

4.2.1 Approach 1: CalculateAndSaveRecommendation

Send user-input and rendered UI data to the backend, calculate the recommendation, record it for audit and return it.

Pros:

1. Single source of truth for business logic
2. Better audit trail with complete calculation history in logs
3. Reduced risk of client-side manipulation
4. Simplified frontend implementation
5. Easier testing and validation of logic

Cons:

1. Higher backend load
2. Increased API response time
3. More complex error handling needed
4. Less responsive UI experience

4.2.2 Approach 2: SaveRecommendation (Recommended)

Calculate the recommendation on the UI and send the user-input, rendered UI data and the recommendation to the backend to record it for audit.

Pros:

1. Faster user experience since the recommendation will be rendered instantly
2. No wait time for backend calculations
3. Lower latency impact
4. Decoupled calculation and storage operations since the calculation highly relies on frequently changing user input

Cons:

1. Potential for client-side manipulation
2. More complex frontend testing needed
3. Harder to debug calculation issues

4.3 RECOMMENDATION AUDIT AND STORAGE

This will be a comprehensive tracking system that captures and stores all partner placement evaluation decisions made through HUGO - including which partner to recruit and how many partners in a recruitment zone are required. It will record user inputs, calculation parameters, and final recommendations to maintain a historical record of recruitment decisions.

This will help in analyzing the effectiveness of recommendations, understanding decision patterns, and reporting HUGO's performance.

Ingestion

The record will be created when a user evaluates a potential partner location through the Partner Placement Evaluator. When the user submits their inputs (target ADV, delivery radius, selected jurisdiction), the system calculates various metrics including forecasted ADV, and attainable volumes.

These calculations, along with the user's input parameters and the final recommendation, will be captured in a single audit record.

The system will support querying of audit records by marketplace, time range, recommendation status, and lead ID, enabling analysis of historical decisions and their outcomes.

4.3.1 Query Parameters

- Country Code & Recruitment Zone Type (ZIP, ZAP, or Jurisdiction)
- Country Code & Recruitment Zone Type & Recruitment Zone ID (partition id)

4.3.2 Data Storage**Approach 1: DynamoDB (Recommended)**

Primary Key - CountryCode#RecruitmentZone#ZoneId

Sort Key - CreationEpoch#AuditId

Pros:

- Efficient queries for both query patterns (point lookups + range scans).
- Built-in time-ordering (because sort key includes timestamp).
- Low latency and high scalability for integrating with UI in the future
- Pay-per-use pricing - hence no idle cost

Cons:

- No query flexibility unless GSIs are added, hence limiting the fields through which the records can be queried

Approach 2: Serverless Relational Database - Amazon Aurora

Composite Primary Key - (CountryCode, RecruitmentZone, ZoneId, Timestamp)

Pros:

- Query flexibility for all combinations of fields
- Rich SQL features like joins, aggregations make it great for reporting - if needed on UI - with direct support for BI dashboard

Cons:

- Pricing is done for provisioned instances and hence will have idle cost
- Higher latency compared to DynamoDB (>100 ms)

Approach 3: S3 + Athena

The data will be stored in S3 and will be queried through Athena, hence supporting all filtering options which SQL provides.

Pros:

- Ideal for ad-hoc queries and report generation
- Lower storage cost compared to other approaches

Cons:

- High latency for API integration
- Cost scales if Athena queries increase

4.3.3 Data Model

Primary and Sort key closed in the section above

- Country Code - String
- Recruitment Zone - String
- Recruitment Zone Id - String
- AuditId (UUID) - String
- Timestamp (Epoch) - Number
- MGRS-level attainable volume - Map<String, Double>
- Active Partners - List<String>
- Onboarding Partners - List<String>
- Leads - List<String>
- Benched Partners - List<String>
- Recruited Partners - List<String> *Note - Name is misleading and will be changed. Recruitment is simulated for these partners during calculation.*
- Waitlisted Partners - List<String>
- UserId (String)

- TargetAttainment - Number
- Work Days - List<String>
- Cycle - String
- BAU Target ADV - Number
- Peak Target ADV - Number
- DeliveryRadius - Number
- DeliveryRadius Unit - String

Fields present if recommendation generated for a selected lead:

- LeadId - String
- Recommendation (Go/NoGo) - String
- ForecastedBAUAdv - Number
- ForecastedPeakAdv - Number
- Lead Delivery Radius - Number
- Lead Delivery Radius Unit - String
- Geocodes - Object with latitude and longitude

5 Recrutable Area Calculation

5.1 DATA REQUIRED

Target Attainment % (Recruitment Area) 40 Must be Whole Integer	Working Days All	Cycle Cycle 1	☒ Peak Up ADV	BAU Target ADV 35 Must be Whole Integer	Peak Target ADV 80 Must be Whole Integer	Delivery Radius 300 M Must be in 50 M Increments
Active Partners All	Onboarding Partners All	Prospective Partners (Salesforce)	Benched Partners	Recruited Partners 1114883830, 1126588959	Wait Listed Partners	

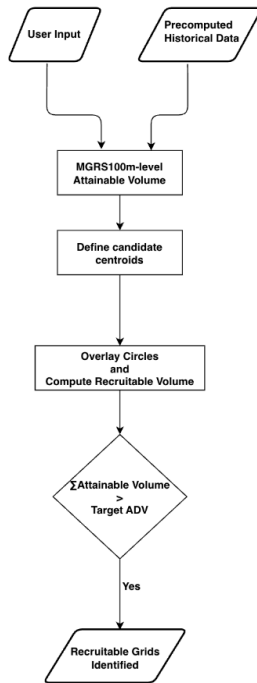
User Input

1. Target Attainment %
2. Work Days
3. Cycle
4. Target ADV (Average Daily Volume)
5. Delivery Radius (in meters)
6. Number of active partners
7. Number of benched partners
8. Number of prospective partners
9. ZIP Code / ZAP Code / Jurisdiction Id - Recruitment Zone & ID

Existing Data

1. Historical average volume at MGRS 100m grid level for the recruitment zone (segmented by cycle and work days)

5.2 STEPS TO FIND RECRUITABLE AREA



Step 1: Preprocess Data

- Find MGRS grids in the selected recruitment zone
- Filter historical volume data for the given ZIP code, cycle, and work days for each MGRS grids

Step 2: Calculate MGRS-Level Attainable Volume

- Formula:

$$\text{Attainable Volume (Hub)} = [\{\text{Total Package Volume}\} * \{\text{Target Attainment \%}\}] - \text{Hub Volume}$$

Step 3: Define Candidate Circle Centers

- Options:
 - Center of each MGRS grid
 - Intersections of grid lines
 - Midpoints of grid borders
- This is done to limit the number of possible partner positions we are considering.

Step 4: Overlay Circles and Compute Recrutable Volume

- Draw a circle with the input delivery radius around each centroid.
- Identify MGRS grids intersecting with the circle.
- For partial overlaps, weight the grid's attainable volume by the % of grid area covered by the circle.
- Sum weighted attainable volumes inside the circle.

Step 5: Check Against Target ADV

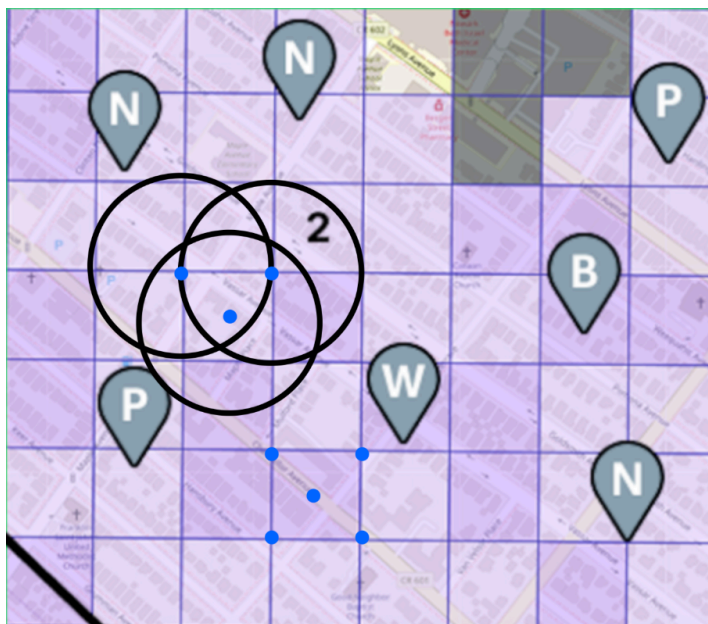
- If the summed attainable volume $\geq$ Target ADV, mark the grids in the circled area as recruitable.

Edge cases to consider:

- Circles partially outside ZIP code boundaries → Any circles with center inside the ZIP will be considered, along with the MGRS grids covered by it.
- Grids with very low or zero historical volume → If such grids are included in the final summation of attainable volumes, they will be displayed as recruitable grids. Heatmap on the UI will help users easily differentiate them from high-volume grids.
- Delivery radius smaller than grid size (100m) - for example, 50m → The weighted calculation of the attainable volume based on area coverage will provide an approximation. We will evaluate the accuracy of such cases when we have more data.

5.3 PROPOSED ALGORITHMS

The below diagram represents possible centroids on the grid in blue and the circles of 100m radius drawn from some of the centroids as a visual representation of the algorithm.



5.3.1 Algorithm 1: Brute Force Scan

Idea

- Place a circle of radius R around each candidate centroid.
- For each circle, check **all MGRS grids** in the ZIP polygon.
- Compute weighted volume:

$$\text{Weighted Volume} = \text{Attainable Volume of Grid} \times \text{Fraction of Grid Covered by Circle}$$

- Sum volumes inside the circle and compare with Target ADV.
- If $\text{sum} \geq \text{Target ADV}$, consider all grids in the circle area as recruitable.

Pros:

- Simplest approach for smaller datasets
- Simulates the exact assumptions and calculations made while designing the product

Cons:

- Does not scale well for larger recruitment zones and more centroids to consider on the map
- High complexity - $O(N*M)$, where N is the number of centroids and M is the number of grids

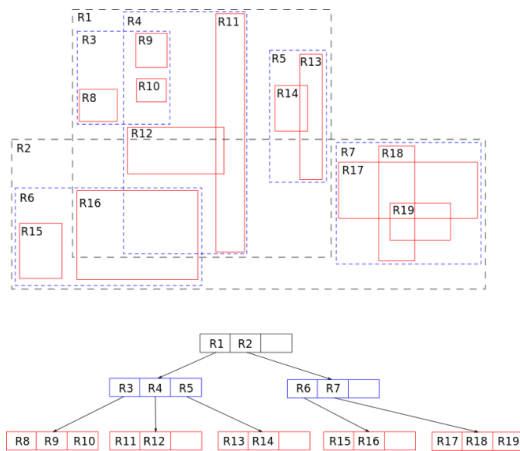
5.3.2 Algorithm 2: Spatial Indexing using R-Tree (Recommended)

Spatial Indexing

A spatial index is a data structure that allows optimizing queries by organizing multi-dimensional spatial data like points, lines and polygons into a tree-like structure instead of checking all of them.

R-Tree

It's a spatial index for polygons where each node of the tree contains bounding boxes. It's self balancing and has handling for overlapping spatial regions for efficient queries.



Idea

- Build an R-tree from all MGRS grids.
- For each centroid circle, **query the spatial index** to get only grids that intersect the circle.
- Compute weighted volume as in Algorithm 1.
- Precise check is supported for each grids intersecting the candidate circles, resulting in a more accurate approximation

Pros:

- Much faster than brute force, scales to large ZIPs.
- Better complexity of $O(N * \log M)$

Cons:

- More complex, requires indexing libraries.

5.3.3 Algorithm 3: MGRS Grid Heatmap Convolution

Convolution

In image processing, convolution is an operation that involves sliding a small matrix (called a kernel) to an image and

computing, at each position, a weighted sum of the overlapping values, hence producing a transformed output that emphasizes patterns such as edges, blurs, or other features.

For our use case,

- image is the attainable volume heatmap of MGRS100m grids
- kernel is a circular matrix defined by 1s and 0s - 1s create the circle
- result of convolution will give us the sum of volumes inside each possible circle positions

Idea

- Convert MGRS grid attainable volumes to a 2D matrix (heatmap).
- Create circular kernel with radius = input delivery radius.
- Convolve kernel with heatmap → matrix of total attainable volumes per possible centroid.
- Threshold by Target ADV to find recruitable centroids.

Pros

- Fast and easy to implement for grids since these are uniform polygons
- Avoids explicit per-grid intersection checks
- Can work in parallel for different centroids
- Highly extensible for predictive analytics using ML, although we don't foresee a use-case for predictive analysis to be done by Hugo.

Cons

- Loses exact polygon precision where kernel approximation may under- or over-count partial coverage
- The above loss results in the calculated approximation to be less accurate than approach 2 where the intersect

5.3 IMPLEMENTATION

The proposed algorithms rely heavily on in-build Python components like shapely, numpy, scipy, and R-Tree, hence requiring the computation to be done via Python instead of UI.

Approach 1: Python-Lambda invoked through API Gateway by Frontend (Recommended)

Pros:

- Leverages Python's highly reliable libraries for geometry and numerical computation.
- Enhanceable to support more accurate computations using Python
- Easy to integrate ML or predictive models in the future.
- Frontend gets ready-to-use results via API.

Cons:

- Adds an extra infrastructure component (Lambda), hence increasing OE load
- Slightly higher cost due to an additional Lambda

Approach 2: Use STR-Tree instead of R-Tree in existing Java API Lambda

STR-Tree is also a type of spatial index used to organize geometric objects. Conceptually, it's a variant of an R-Tree, optimized for bulk loading and query speed. Using an STR-Tree instead of a traditional R-Tree does not reduce accuracy, because both store the same geometric envelopes; the difference is only in how the tree is built and traversed, not in the

spatial calculations themselves. This means all intersection checks and coverage fractions will produce the same results.

Pros:

- Fully integrated into existing Java backend, hence simpler maintenance.
- Efficiently handles spatial queries for large datasets.

Cons:

- Geometry computations are more verbose than Python, thus requiring more effort to setup.
- Less convenient if we want to add learning models later.
- Harder to prototype complex numerical operations compared to Python.

6 Low Level Design (WIP = Do not Review)

6.1 SAVERECOMMENDATION API

6.1.1 Input Model

```
{
  "requester": "alias",
  "country": "US",
  "leadRecordId": "313131313",
  "recruitmentConfig": {
    "selectedWeek": 30,
    "selectedYear": 2025,
    "targetAttainmentPercentage": 40,
    "deliveryRadius": 300,
    "deliveryRadiusUnit": "M",
    "recruitmentTarget": "BAU/PEAK",
    "targetAdv": 35,
    "selectedCycle": "Cycle1",
    "workDays": ["MONDAY", "TUESDAY", "WEDNESDAY", "THURSDAY", "FRIDAY"]
  },
  "recruitmentArea": {
    "areaType": "ZIPCODE",
    "partitionIds": [],
    "boundingBox": {
      "northeast": {
        "latitude": 12.34567,
        "longitude": -45.67890
      },
      "southwest": {
        "latitude": 12.34567,
        "longitude": -45.67890
      }
    }
  },
  "partnerSelections": {
    "included": [],
    "excluded": []
  }
}
```

6.1.2 Output Model

```
{
  "statusCode": "200"
}
```

6.1.3 Sequence Diagram

6.1.4 Class Diagram

6.1.5 Exception Handling

6 Data Pipeline for Metrics and Reporting

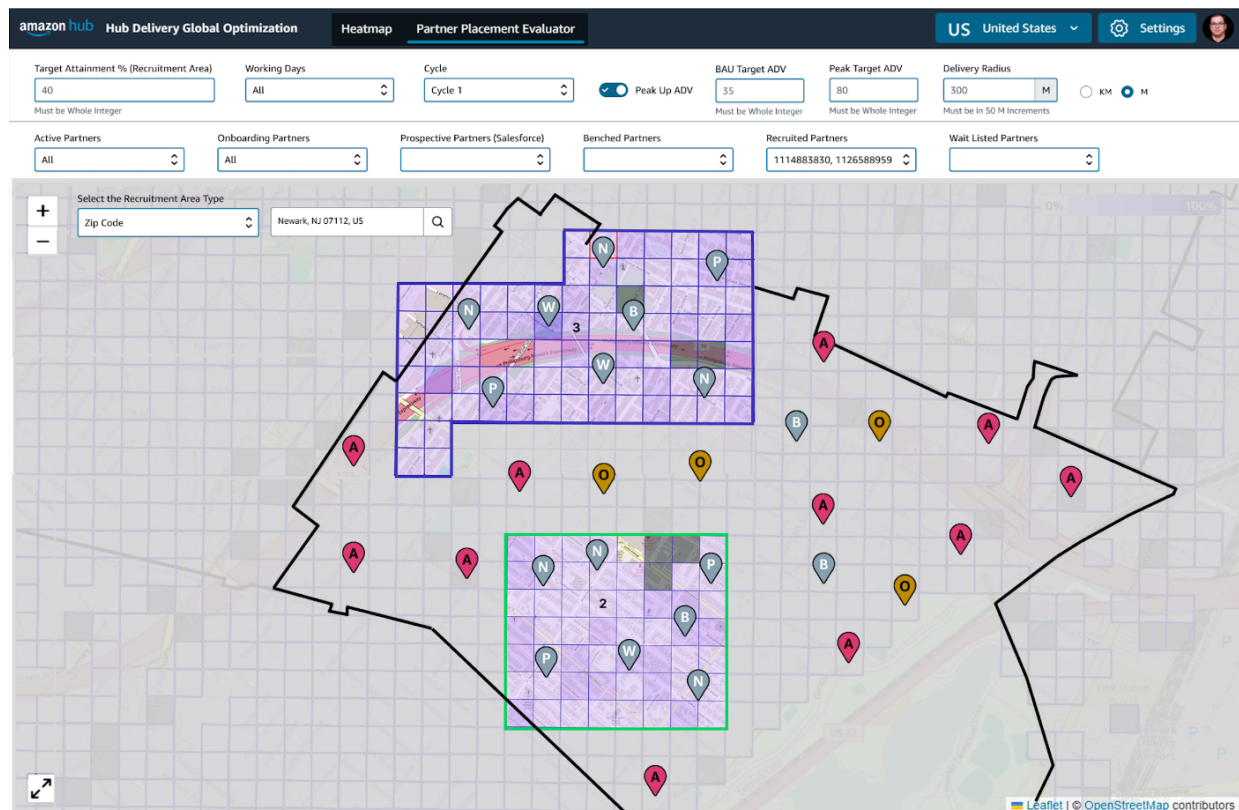
7 Appendices

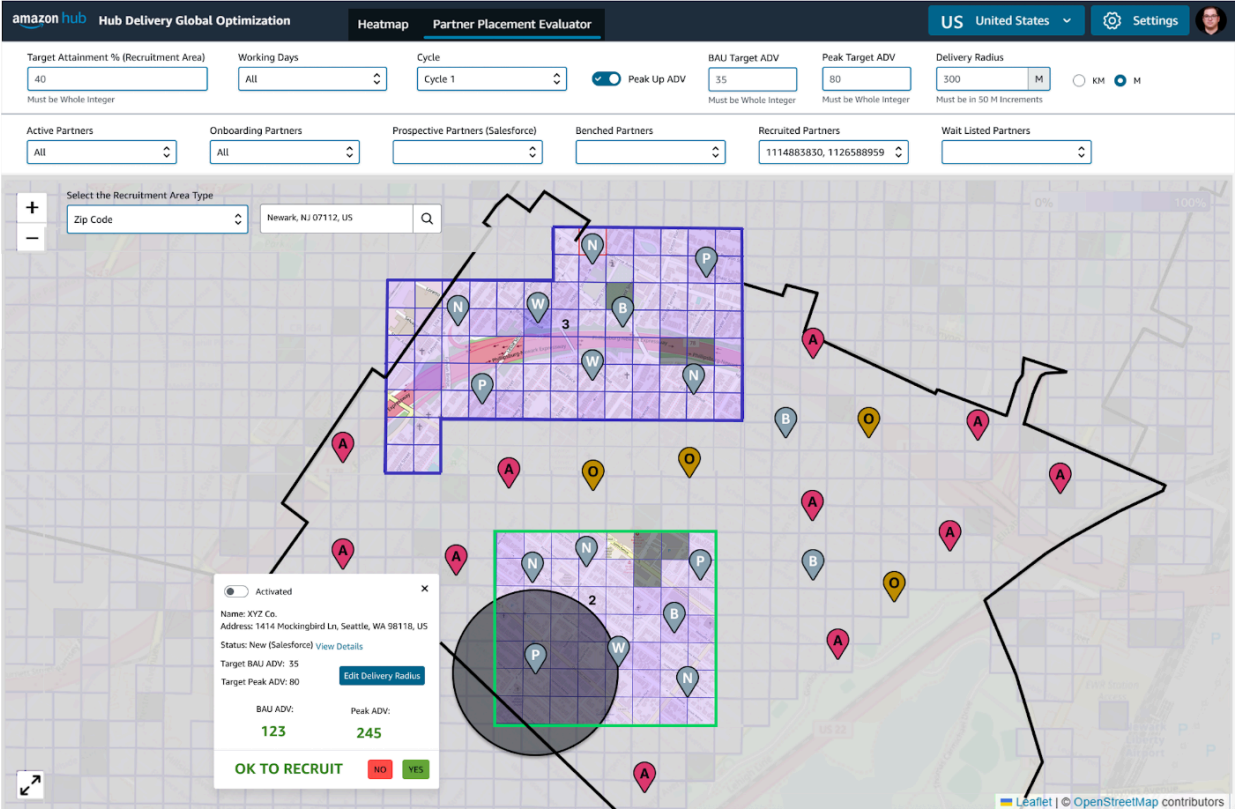
7.1 TENETS

7.2 FAQ

7.3 REFERENCES

7.4 UI MOCKS





Select the Recruitment Area Type

Zip Code

Enter Zip Code

Q

Zip Code

Zip Code

Zip Code

Exclusive Hub Delivery Jurisdiction

Select the Recruitment Area Type

Zip Code

Enter Zip Code

Q

Zip Code

Enter Zip Code

Q

Zip Code

Enter Zip Code

Q

Exclusive Hub Delivery Jurisdiction

Enter JAS ID

Q

Select the Recruitment Area Type

Zip Code

Newark, NJ 07112, US

Q

Zip Code

Z03-US-EWR-44020

Q

Exclusive Hub Delivery Jurisdiction

Hub_Panama_City_32401

Q

Target Attainment % (Recruitment Area)

40

Must be Whole Integer

Target Attainment % (Recruitment Area)

40.5

Must be Whole Integer (i.e. - 20)

Working Days

All

All

Monday

Tuesday

Wednesday

Thursday

Friday

Saturday

Sunday

Working Days

All

All

Monday

Tuesday

Wednesday

Thursday

Friday

Saturday

Sunday

Working Days

M, TU, W, TH, F, SA

All

Monday

Tuesday

Wednesday

Thursday

Friday

Saturday

Sunday

Cycle

Cycle 1

Cycle 1

Cycle 1

Cycle 2

Cycle 3

Cycle

Cycle 1

Cycle 1

Cycle 2

Cycle 3

BAU Target ADV

Peak Up ADV

35

Must be Whole Integer

Peak Target ADV

80

Must be Whole Integer

BAU Target ADV

Peak Up ADV

35

Must be Whole Integer

Delivery Radius

1.5

KM

KM

M

Must be in 0.5 KM increments

Delivery Radius

300

M

KM

M

Must be in 50 M increments

111

Select all results

1022579111

111003f4-fba-4d53-a93...

1110921770

1112692076

1114883830

1115484672

1115732724

1117167213

1117830088

1118331132

1 - 120

SEARCH

Show selected

APPLY

1114883830, 1126588959,....

Search value

Q

SEARCH

Selected values

1114883830

1126588959

2288525409

Show selected

APPLY

1114883830, 1126588959

Search value

Q

SEARCH

Selected values

1114883830

1126588959

Show selected

APPLY

1114883830

Search value

Q

SEARCH

Selected values

1114883830

Show selected

APPLY

1114883830

Search value

Q

SEARCH

Show selected

APPLY

Search value

Q

SEARCH

Show selected

APPLY

All

De-selected Partner

Recruitment Evaluation

Recruitment Decision

Active Partner (Admiral)

Onboarding Partner (Admiral)

Benched Partner (Admiral)

BONAP

☒ Active

Name: XYZ Co.
 Address: 1414 Mockingbird Ln, Seattle, WA 98118, US
 Status: New (Salesforce) [View Details](#)

Name: XYZ Co.
Address: 1414 Mockingbird Ln, Seattle, WA 98118, US
Status: Active (Admiral) [View Details](#)

  Active 

Name: XYZ Co.
Address: 1414 Mockingbird Ln, Seattle, WA 98118, US
Status: Profile Submitted (Admin) [View Details](#)

  Active 

Name: XYZ Co.
Address: 1414 Mockingbird Ln, Seattle, WA 98118, US
Status: Inactive (Admiral) [View Details](#)

B

O

P

W

N

Active

X

Name: XYZ Co.

Address: 1414 Mockingbird Ln, Seattle, WA 98116, US

Status: New (Salesforce) [View Details](#)

Calculate ADV



☒ Active

Name: XYZ Co.
 Address: 1414 Mockingbird Ln, Seattle, WA 98118, US
 Status: Profile Submitted (Admin) [View Details](#)

[Calculate ADN](#)

B

Active

X

Name: XYZ Co.

Address: 1414 Mockingbird Ln, Seattle, WA 98118, US

Status: Inactive (Admiral) [View Details](#)

Calculate ADV

B

O

P

N

Active

X

Name: XYZ Co.

Address: 1414 Mockingbird Ln, Seattle, WA 98118, US

Status: New [\(Salesforce\)](#) [View Details](#)

Target BAU ADV: 55

Target Peak ADV: 80

BAU ADV:

123

Peak ADV:

245

OK TO RECRUIT

NO

YES

Edit Delivery Radius

B

O

P

W

N

Active

×

Name: XYZ Co.

Address: 1414 Mockingbird Ln, Seattle, WA 98115, US

Status: New (Salesforce) [View Details](#)

Target (BAU) ADV: 35

Target Peak ADV: 80

BAU ADV:

40

Peak ADV:

70

MOVE TO WAIT LIST

Yes

No

Edit Delivery Radius

Delivery Radius ☒ M ☐ KM

Must be in 50 M Increments

Latitude

Longitude

Recruited Partner

☒ Active

Name: XYZ Co.
Address: 1414 Mecklingbird Ln, Seattle, WA 98115, US
Status: New [Salesforce] [View Details](#)

Target BAU ADV: \$5
Target Peak ADV: \$0

BAU ADV:	Peak ADV:
123	245

Recruited By: mschuler Date: July 5, 2025 [Reset](#)

Waitlisted Partner

☐ Active

Name: XYZ Co.
 Address: 1414 Mockingbird Ln, Seattle, WA 98115, US
 Start-up: Now [\[Salesforce\]](#) [view Details](#)
 Target BAU ADV: 35
 Target Peak ADV: 80

BAU ADV: **40**

Peak ADV: **70**

Wait Listed

By: [michael](#)

Date: July 5, 2025

[Book](#)